

# FORKAUDIT: OWNERSHIP-TRACE VALIDATION FOR HYBRID LLMs BEYOND OUTPUT EQUALITY

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## ABSTRACT

Forking many continuations from one prefix creates an ownership question that output equality cannot resolve: agreeing branches may still alias mutable recurrent state or share a defect. We present FORKAUDIT, a phase-aware, fail-closed ownership-trace validator for hybrid KV and recurrent state. It binds lifecycle, storage, tail-COW, transition, call, and semantic receipts, and it separates evidence coverage from replay verdicts: a missing mandatory record leaves the corresponding target open.

On one fixed Qwen3.5-35B-A3B/H20-3e stack, six of seven audit obligations have complete evidence coverage and pass; dispatch remains partial because the trace binds the Python call but not per-call compiled dispatch. Across 96 KV-by-GDN ownership configurations, all registered semantic observables match and transition receipts establish private mutable GDN state. In the  $N = 32$  materialized-GDN arm, shared-document rather than full-copy KV lowers final allocated delta above the frozen post-priming baseline from 4.901 to 2.229 GiB (54.5%). Separately, in a same-checkpoint eight-item deployment cohort, CoMem Q8 reduces median retained-document tensor payload from 140.34 to 15.89 MiB/document (88.68%) versus full-prefix Q16 caching at the same measured mean F1 of 39.14; a per-layer mixed variant retains 9.74 MiB/document (93.06% less) with a mean-F1 delta of  $-0.022$  points. This is a Store-F1 comparison, not total process memory or capacity. All nine designed all-gates-on mutations and five designer-executor-separated mutations reach their frozen predicates under passing matched controls. One retrospectively reproduced alias preserves tokens, full logits, terminal request GDN, and logical KV while corrupting the persistent base; the repaired path is storage-clean in all eight frozen cells. The audit supports fixed-stack, non-adversarial offline CI at declared observation points, not runtime attestation; faithful producer enumeration remains in the TCB.

## 1 INTRODUCTION

Best-of- $N$  decoding, agents, tree search, and document QA fork many continuations from one prefix. Paged KV caches, prefix trees, and scheduling exploit that structure (Kwon et al., 2023; Zheng et al., 2024; Gim et al., 2024; Srivatsa et al., 2025). Hybrid models add mutable recurrent or linear-attention state to read-only KV, so sharing also requires an explicit request-local recurrent view (Yang et al., 2025; Lenz et al., 2025; Pan et al., 2025).

Output equality is only relational: branches can agree while aliasing state, and compared implementations can share a defect. An ownership audit must also witness prefix immutability, phase-conditioned request/base and request/peer relations, tail COW, transition-time rebinding, and call provenance. Likewise, logical payload, cache pages, allocator peaks, process memory, and admitted capacity are distinct memory denominators.

FORKAUDIT makes this evidence explicit. It partitions document and request state, records page reservations and recurrent rebinding, fingerprints the Python attention callable, and replays relational, physical-storage, and selected numerical checks over a  $2 \times 2$  ownership factorial. The phase-indexed contract targets accidental state-management regressions in instrumented offline CI; its capture/replay path is an explicit TCB (Section 3).

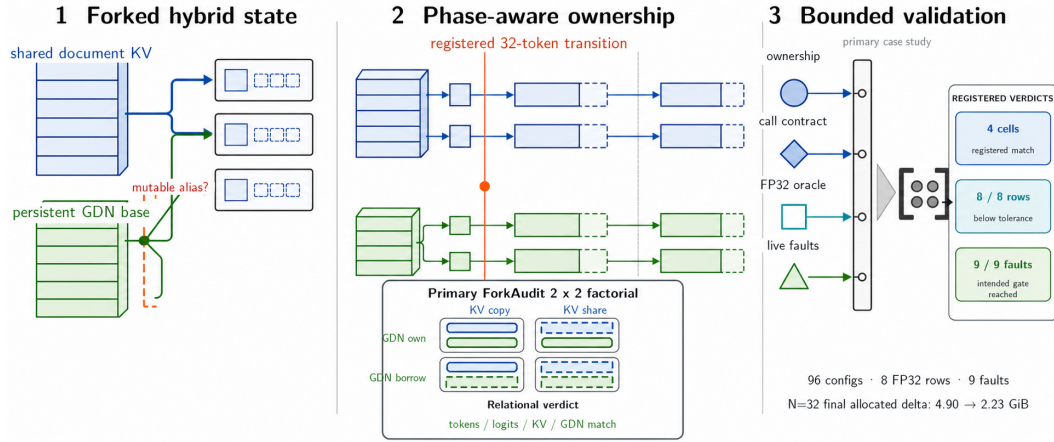


Figure 1: FORKAUDIT in one view. A common prefix forks into requests with distinct KV and recurrent-state lifetimes. Four ownership cells feed private append and recurrent-rebinding checks. The right-hand audit rail maps ownership/call receipts, selected FP32 attention checks, and positive-control faults into registered verdicts. The callouts summarize 96 configurations, selected numerical checks, nine primary live faults, and the  $N = 32$  materialized-GDN endpoint: shared-document KV lowers post-priming final allocated delta from 4.90 to 2.23 GiB (54.5%) versus full-copy KV. This is a protocol schematic; Section 5 provides quantitative authority.

The core contribution is an executable ownership contract, not a new cache policy or production runtime. A target can pass only when phase-complete evidence, pointer-free ownership predicates, and semantic relations hold for both KV and recurrent state. Output relations alone can miss aliasing; storage checks without lifecycle evidence can miss transition-time rebinding. On one fixed stack, the primary evidence track evaluates this contract; a separate same-checkpoint cohort measures retained-document Store versus F1 and is not FORKAUDIT validation evidence:

- **Fail-closed hybrid-state contract.** Mandatory records cover document KV, private append, GDN bases and rebinding, tail COW, and call contracts; missing records prevent a pass, and coverage remains distinct from the replay verdict.
- **Replayable ownership evidence.** Across 96 configurations, semantic invariance is coupled to registered-point physical isolation. Process-separated receivers derive 1,080 GDN descriptors and 96,660 relations from opaque capture IDs and CUDA-IPC handles while slot enumeration stays in the TCB. In the  $N = 32$  materialized-GDN arm, shared-document KV reduces the post-priming allocator delta 54.5% versus full-copy KV while the registered observables remain canonically equal.
- **Separate deployment Store-F1 result.** On the same checkpoint/runtime and eight-item Qasper/2WikiMQA slice, the per-layer CoMem variant retains 9.74 versus 140.34 MiB/document for full-prefix Q16 (only 6.94% as much; 93.06% less) while measured mean F1 changes by  $-0.022$  points. Q8 retains 15.89 MiB (11.32% as much; 88.68% less) with a measured mean-F1 delta of 0.000. Full-prefix Q16 remains faster; Table 3 defines the retained-Store denominator.
- **Falsification-oriented evaluation.** Captured-boundary numerical checks reject 44 seeded wrong-operator controls; nine designed all-gates-on mutations and five designer-executor-separated mutations reach their frozen predicates. A separate hash-bound case study reproduces and repairs one historical no-injection alias defect. Outcomes are reported per frozen fault or cell, not as a population rate.

The evaluation uses batch-one, sequential round-major calls on one fixed Qwen3.5/H20 stack; Section 6 states its boundary.

## 2 BACKGROUND AND RELATED WORK

Prior work develops prefix sharing, hybrid-state caching, cache compression, and relation-based testing. Our question is narrower: what recorded evidence establishes ownership across coexisting immutable and mutable state?

**Paged and prefix-aware inference.** PagedAttention, Prompt Cache, RadixAttention, ChunkAttention, Hydragen, and Preble establish block sharing, prefix reuse, shared-prefix computation, and scheduling (Kwon et al., 2023; Gim et al., 2024; Zheng et al., 2024; Ye et al., 2024; Juravsky et al., 2024; Srivatsa et al., 2025). We instead ask what execution evidence demonstrates isolation across coexisting state types.

**Hybrid state and cache management.** Gated Delta Networks maintain mutable compact state (Yang et al., 2025); Jamba, Marconi, and HYPIC address hybrid architectures or reusable hybrid prefixes (Lenz et al., 2025; Pan et al., 2025; Liu et al., 2026c), while Palu, SqueezeAttention, and HeadKV compress or budget KV (Chang et al., 2025; Wang et al., 2025; Fu et al., 2025). FORKAUDIT validates an ownership boundary rather than replacing those policies; Appendix Table 13 separates published numbers from our controls.

**Persistent intermediate residual reuse.** CoMem persists a depth- $j$  residual per document token (Liu et al., 2026a;b), motivating our persistent document object; its bounded deployment slice supplies a retained-memory-quality measurement, not a new selector or quality-latency claim.

**Evaluation and metamorphic testing.** SCBench emphasizes cache lifecycles (Li et al., 2025); metamorphic testing, MT-DLComp, and CRADLE motivate preserved relations and independent checks (Chen et al., 2018; Xiao et al., 2022; Pham et al., 2019); DNNMem separates memory denominators (Gao et al., 2020). Appendix Table 10 gives our component-wise boundary.

## 3 THE FORKAUDIT CONTRACT AND CONDITIONAL ASSURANCE BOUNDARY

Consider a document prefix  $D$  and resident requests  $r \in \{1, \dots, N\}$ . Let  $\mathcal{A}$  be the full-attention layers and  $\mathcal{G}$  the recurrent linear-attention layers. For  $\ell \in \mathcal{A}$ , the state is a read-only set of document pages  $K_\ell^D$  plus a request-private reservation  $K_\ell^r$ . For  $\ell \in \mathcal{G}$ , it is a persistent prefix base  $G_\ell^D$  plus the request-local recurrent state  $G_\ell^r$ . A logical fork is

$$\mathcal{F}_r(D) = (\{K_\ell^D, K_\ell^r\}_{\ell \in \mathcal{A}}, \{G_\ell^D, G_\ell^r\}_{\ell \in \mathcal{G}}). \quad (1)$$

**Threat model and assurance boundary.** Here “audit” means an offline, non-adversarial contract check for debugging and regression testing, not security attestation. The instrumented state-management path may allocate, alias, append, rebind, or dispatch incorrectly, but is not assumed to coherently forge, suppress, or relabel observations. The capture/replay path, mandatory-slot schema, and framework tensor/storage semantics form an explicit TCB, including faithful producer enumeration at every registered phase. Under this TCB, a pass establishes that the registered predicates held at the captured slots and times; it does not validate the TCB or protect against coherent omission/fabrication, malicious-runtime behavior, OS/driver faults, or unobserved transient writes.

A bounded GDN cohort moves descriptor/relation derivation to `spawn`-created processes. The producer supplies opaque capture IDs and CUDA-IPC tensor handles for the frozen slot set; receivers derive descriptors and all-pairs relations and emit no verdict. They receive neither candidate rows/verdicts nor live phase/policy/completion labels, while producer slot enumeration, PyTorch/CUDA-IPC semantics, and paused capture remain in the TCB. We therefore separate mandatory-trace *coverage* from replay *verdicts*. A *target* is an audit obligation and a *receipt* one evidence item; coverage reports whether mandatory receipts are present, whereas a verdict reports replay. A *gate* is the first live predicate that stops a faulted execution. “Registered” means fixed by the frozen protocol, not independently observed.

**Audit targets.** Accordingly, the seven audit targets may have complete, partial, or open coverage:

1. **Frozen identity.** Model, data, query bank, layer geometry, kernel descriptor, and protocol hashes are fixed before execution.
2. **Prefix immutability.**  $K_\ell^D$  and  $G_\ell^D$  have identical digests before and after all continuations.
3. **Private ownership.** KV reservations of distinct shared-document requests are disjoint in the common arena; independent materialized arenas have disjoint storage. At setup, a GDN request either owns a materialized base or holds an explicitly read-only persistent-base alias. At the primary factorial’s registered transition, all 60 mutable tensors must be disjoint from the base and every peer.
4. **Tail-safe append.** If the last document page is partial, its valid tokens are copied once to a private page before a request appends.
5. **Dispatch provenance.** Both controls invoke the same recorded Python callable and query/mask/position/append contracts; fallback and full-KV concatenation are forbidden. Compiled binaries and autotuning are not fingerprinted.
6. **Cross-arm equivalence.** For the same query and schedule, the materialized and shared ownership arms must have exactly equal generated tokens, full-vocabulary logits, final logical KV, and request recurrent state.
7. **Cross- $N$  prefix consistency.** For each query observed at two or more fan-outs, its result at every larger fan-out must match its smallest- $N$  result.

Targets 2–5 are pre-output trace obligations; targets 6–7 are relational semantic checks, so matching outputs is insufficient. The nine controls stop at frozen first gates before post-injection semantics: bounded localization, not unseen-fault coverage. Separate executions suppress only the named gate and record the next outcome under frozen precedence; executions are not pooled into a rate. The retrospective historical case is likewise a post-discovery reproduction of one defect mechanism, not a blind-fault or defect-population study.

**Conditional trace-validity statement.** Let  $\Sigma$  be the frozen case specification,  $E$  an execution, and  $\tau = C_\Sigma(E)$  the producer trace. For target  $i$ , let  $M_i(\Sigma)$  be its mandatory record slots and  $\Phi_i$  its registered replay predicates. Coverage is *complete* only when every slot in  $M_i(\Sigma)$  has exactly one schema-valid, byte-bound record. We define

$$\text{Pass}_i(\tau) \iff [\text{Coverage}_i(\tau) = \text{complete}] \wedge \text{Bind}_\Sigma(\tau) \wedge \bigwedge_{\phi \in \Phi_i} \phi(\tau). \quad (2)$$

If capture is honest and mandatory-event-complete, and manifest binding, digests, and replay are correct,  $\text{Pass}_i(\tau)$  implies that every registered predicate held at its captured observation point. A missing or modified mandatory record, or a represented predicate violation, cannot produce a pass. This statement is trace-relative: it provides no conclusion about coherent omission or fabrication, transient writes restored between observations, unrecorded compiled dispatch, common-mode semantics, or unseen-fault error rates.

The out-of-process cohort is a separate supporting check on a bounded subset of GDN storage predicates; it does not change this conditional statement for the complete primary-factorial producer trace.

**Minimal adoption workflow and decision rule (normative).** FORKAUDIT (1) freezes code/model/data/protocol/runtime roles before outputs; (2) captures setup, transition, and final events with canonical digests, ranges, ordered receipts, outputs, and allocator snapshots; (3) replays every applicable predicate, labeling coverage complete, partial, or open separately from pass/fail/not-evaluated; and (4), when sensitivity is claimed, freezes the target-to-gate map and retains all outcomes. Here frozen identity is complete and passing. For dispatch, the recorded Python-call predicate passes, but target coverage remains partial because compiled dispatch and autotuning are not bound. Appendix D and Table 17 give the dependency matrix, pointer-free schema, and record obligations.

Throughout, “canonical equality” means equality of the archived canonical serialization: contiguous CPU-FP32 bytes with bound shape/dtype, or the specified logical-KV/GDN digest. It does not imply correctness of unobserved tensors or end-to-end semantics.

**Selected numerical support.** Targets 6–7 expose layout- or  $N$ -dependent changes in recorded observables. The factorial also captures one statically selected rank-specific attention row’s post-RoPE

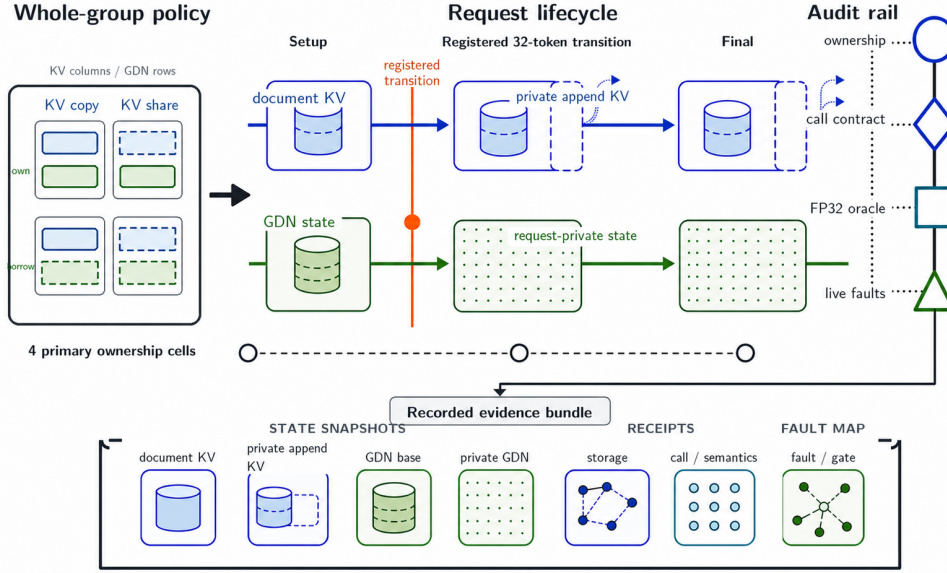


Figure 2: ForkAudit’s ownership factorial. The two left axes choose one KV and one GDN setup policy for the request group: append tails are always private, and a borrowed GDN base becomes private at the registered 32-token transition. The dashed lifecycle spine marks setup, that transition, and final capture. Left-to-right, the bracketed bottom row records document KV, private append, persistent GDN, request-private GDN, storage binding, call/semantic, and fault-to-gate evidence. Offline replay validates that bundle and reports coverage separately from each verdict. This mapped schematic is not experimental evidence.

query, K/V, positions, mask, and scale for dense IEEE-FP32 comparison. A candidate-import-free NumPy FP32 oracle evaluates four frozen GDN transitions after native BF16 q/k normalization. Preregistration binds that boundary,  $1/\sqrt{128}$  query scale, rows, tolerances, and seeded faults. These checks cover selected operator boundaries, not upstream construction, normalization, every GDN layer, or the end-to-end model.

Figure 2 maps the four primary-factorial cells and their lifecycle; allocator and witness cells are rebuilt separately so guards do not retain tensors during measurement.

## 4 TRACE-VALIDATION PROTOCOL

### 4.1 FROZEN CASE-STUDY GEOMETRY

Primary feasibility uses one fixed Qwen3.5-35B-A3B configuration (Qwen Team, 2026a;b): ten vLLM-0.26 Triton full-attention layers with BF16 (“Q16”) KV and 128-token pages, plus thirty Transformers-torch GDN layers. Eight H20-3e ranks run one book each with batch-one sequential calls on one CUDA stream/rank. The ledger binds the Python callable, not the compiled binary; each GDN digest covers 60 tensors.

Each distinct PG-19 book (Rae et al., 2020) supplies a 4,095-token document and 32-token registered transition. Four cells evaluate  $N \in \{1, 8, 32\}$  and eight argmax outputs; the query plus seven fed-back outputs give 39 state-appended tokens and exercise COW on the 127-token final page. Appendix Table 5 separates all supporting cohorts.

### 4.2 AUDIT SEQUENCE

Each primary-factorial cell binds identity, rebuilds allocator and witness executions, records setup–transition–final ownership, and compares tokens, logits, logical KV, and GDN state. Replay adds fixed-axis projections, one dense-FP32 sidecar per rank, and matched live faults. A separate PDF-

only designer freeze supplies five executor-separated pairs. Aggregation requires every byte binding. Appendix Table 6 states target strength.

#### 4.3 CONTROLS, SCHEDULES, AND ORACLES

The primary factorial uses sequential round-major steps on one CUDA stream per rank. Full logits use shape/dtype-bound CPU-FP32 bytes; KV, GDN, bases, calls, storage, and allocator records use typed digests. Detached replay cannot reconstruct unarchived tensors or recapture the producer. A preregistered NumPy oracle checks four GDN transitions at global model-layer indices 0, 10, 20, 38 after native q/k normalization against four wrong recurrences.

Supporting protocols move GDN reconstruction to a spawn-created CUDA-IPC observer and record CUDA-event-bracketed model-call intervals around cancel, scrub, reclaim, and replacement. Appendix Table 5 fixes their geometry and authorization.

#### 4.4 MEMORY DENOMINATORS

We separate logical state, pages, synchronized allocator bytes, and unmeasured service capacity. Deployment “Store” is instead the union of tensor byte ranges owned by a retained document entry; it excludes metadata, pools, NVML deltas, and admission capacity.

#### 4.5 GOVERNANCE AND LIVE MUTATION CHECKS

Pre-output ledgers bind code, model, data, queries, and protocol. Nine fresh matched controls pair with live mutations spanning allocation/binding, COW, GDN aliasing, positions, masks, callable identity, and paged-view type. Each mutant first stops at its predeclared gate, establishing localization rather than a rate; the primary all-gates-on campaign therefore records no post-injection token, logit, logical-KV, or GDN digest.

In a separate preregistered 18-case matrix, only the named gate is suppressed. Frozen precedence then records another FORKAUDIT gate, an allowlisted production assertion, the exact M8 payload sentinel, or completed-horizon token and FP32-logit equality. Unallowlisted exceptions are invalid; the sentinel is not a production detector.

The designer-executor-separated campaign’s designer receives only the prior PDF—not source, evidence, earlier faults, reviews, or state—and freezes five new definitions, loci, payloads, clean gates, precedence, and first predicates before executor preparation or candidate output. Every fault runs all gates; selective reruns and post-output payload tuning are forbidden.

The historical-regression protocol is separate from both mutation campaigns. Each rank builds fresh historical-pre-fix, repaired-borrowed, and materialized  $N = 2$  lanes for one token; the fault executor is blocked and no mutation is requested or applied. Ranks 0–2 reproduce archived input coordinates, while ranks 3–7 are additional frozen inputs rather than defect-population samples.

### 5 RESULTS

#### 5.1 PRIMARY FIXED-STACK TRACE VALIDATION

Targets 1–4 and 6–7 have complete mandatory-trace coverage and satisfy the registered pass criterion for the frozen primary case study. Dispatch provenance (target 5) remains partial because no per-call compiled-binary or autotuning-choice binding exists; only its recorded Python-call predicate passes. Thus the audit reports six complete-and-pass targets and one partial target rather than upgrading missing dispatch evidence.

#### 5.2 OWNERSHIP EVIDENCE AND ALLOCATOR RESULT

Across  $8 \times 3 \times 4 = 96$  configurations (192 rebuilt allocator/witness cells), tokens, full logits, final logical KV, final GDN, both fixed-axis projections, and all 288 adjacent-fan-out comparisons match exactly. These semantic relations are paired with physical ownership evidence: borrowed GDN starts

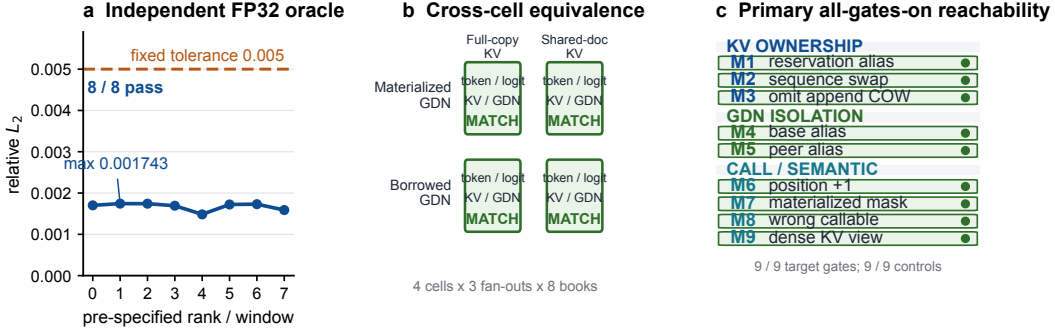


Figure 3: Primary validation. Left: an independently implemented FP32 replay starts from producer-captured post-RoPE inputs, and all selected attention rows beat the pre-fixed tolerance. Center: four ownership cells match on registered observables. Right: a separate primary all-gates-on reference reaches every target gate with passing controls. Together these panels connect selected numerical agreement, cross-cell invariance, and targeted violation localization.

as 60 immutable base aliases/request, materialized setup is disjoint, and both rebind at the registered 32-token transition boundary to 60 request-private tensors disjoint from bases and completed peers. The evaluated ownership and immutability predicates pass at setup, transition, and final capture; dispatch remains partial (Section 5.1). The  $N = 1$  cells directly exercise request-base disjointness.

**Allocator reduction with canonical equality.** With materialized GDN setup at  $N = 32$ , switching from full-copy to shared-document KV reduces final allocated delta from 4.901 to 2.229 GiB, a 54.5% reduction above the frozen post-priming baseline, while the registered observables are canonically equal across the compared ownership arms. Borrowed GDN changes allocation timing rather than the shared-KV end state: its corresponding full-copy and shared-document endpoints are 4.890 and 2.229 GiB, and both shared-document cells finish at 2.229 GiB. Appendix Table 18 reports all four endpoints and the allocator denominator.

**Falsification beyond output equality.** The primary dense-FP32 attention oracle obtains relative- $L_2$  0.001484–0.001743  $< 0.005$  on eight rows. A preregistered expanded numerical sweep uses two distinct PG-19 inputs and checks 20 attention rows spanning all ten full-attention layers (160 query positions) plus 24 GDN rows spanning 12 of 30 recurrent layers (192 token transitions). All 44 clean rows pass: maximum attention relative- $L_2$  is 0.001897; maximum GDN output/state relative- $L_2$  is 0.002073/2.08  $\times 10^{-7}$ . All 20 attention and 24 GDN seeded wrong-operator controls are rejected. The CPU/NumPy FP32 replay imports no candidate code, but begins from producer-captured post-RoPE attention and post-native-q/k-normalization GDN inputs; Appendix Table 23 fixes this captured-boundary scope.

Table 1: Same-system outcomes for nine designed faults. The primary campaign enables all gates; a separate target-gate-suppression campaign suppresses only the named gate. “Same” is exact and “N/O” an earlier stop. Rows are not a detection rate; M8’s injected sentinel is not a production detector.

| Fault / target contract  | All gates on | With target gate suppressed                                          | Token / FP32 logits |
|--------------------------|--------------|----------------------------------------------------------------------|---------------------|
| M1 Reservation ownership | target gate  | completes                                                            | Same / Same         |
| M2 Sequence binding      | target gate  | completes                                                            | Same / Same         |
| M3 Tail COW              | target gate  | other audit: active-block ownership                                  | N/O / N/O           |
| M4 GDN base isolation    | target gate  | other audit: peer isolation                                          | N/O / N/O           |
| M5 GDN peer isolation    | target gate  | completes; logits change ( $L_\infty = 0.75$ , rel. $L_2 = 0.0314$ ) | Same / Changed      |
| M6 Positions             | target gate  | completes                                                            | Same / Same         |
| M7 Mask contract         | target gate  | completes                                                            | Same / Same         |
| M8 Callable identity     | target gate  | injected payload sentinel                                            | N/O / N/O           |
| M9 Paged-KV view         | target gate  | paired-view production assertion                                     | N/O / N/O           |

Table 1 asks what output-only checks miss after the named gate is suppressed. All nine matched clean controls pass and no case is invalid. Among the five semantic-complete mutants, token equality catches 0/5 and exact logits catch 1/5; the frozen FORKAUDIT gates reject all five in their separate all-gates-on executions. The other four cases reach a redundant audit gate, an allowlisted production assertion, or the injected M8 sentinel. Appendix Table 7 gives the full targeted-control boundary.

**Designer-executor-separated fault campaign.** Table 8 reports the separately frozen five-pair campaign. Its faults are held out from executor preparation and output inspection, not from FORKAUDIT’s predicate vocabulary. Every matched clean case passes complete coverage, byte binding, full-horizon execution, and exact allocator restoration; every mutant also completes and is rejected first by its pre-frozen primary predicate. Four of five preserve the surfaced token sequence. Of the four pairs with comparable call cardinality, three preserve every FP32-logit sidecar byte; the duplicate commit intentionally has one extra recorded call. HF01, HF02, and HF05 also preserve terminal logical KV and GDN state. These are five fixed frozen outcomes, not an estimate over a fault population.

**Historical pre-fix regression and repair.** A retrospective reproduction under a preregistered protocol revisits one no-injection integration defect encountered before this protocol: borrowed request, persistent-base, and unadvanced-peer convolution tensors shared storage that the one-token GDN path updated in place. In Table 2, the pre-fix path reaches the expected authenticated binding predicate at all three archived coordinates and five additional frozen inputs. Nevertheless, token, complete FP32 logits, terminal request GDN, and logical KV are exact against a fresh materialized control in 8/8 cells. The repaired path privatizes the 30 convolution states; it is storage-clean and exact to the materialized control in 8/8. A persistent-base content invariant also catches 8/8 historical cells, so this case supports earlier owner/layer/family-specific localization beyond output and terminal-state differentials, not exclusive detection over every conventional invariant.

Table 2: Retrospective reproduction and repair of one historical alias regression. Ranks 0–2 reuse three archived input coordinates; ranks 3–7 are additional frozen inputs, not additional defects or statistical replicates. Semantic columns compare each path with a fresh materialized control. “Base inv.” is within-path persistent-base immutability, retained as a conventional state-invariant catch.

| Coordinates    | Path     | Ownership result  | Token     | FP32 logits | Request GDN | Logical KV | Base inv. |
|----------------|----------|-------------------|-----------|-------------|-------------|------------|-----------|
| Archived (3)   | Pre-fix  | binding gate 3/3  | exact 3/3 | exact 3/3   | exact 3/3   | exact 3/3  | pass 0/3  |
| Archived (3)   | Repaired | storage-clean 3/3 | exact 3/3 | exact 3/3   | exact 3/3   | exact 3/3  | pass 3/3  |
| Additional (5) | Pre-fix  | binding gate 5/5  | exact 5/5 | exact 5/5   | exact 5/5   | exact 5/5  | pass 0/5  |
| Additional (5) | Repaired | storage-clean 5/5 | exact 5/5 | exact 5/5   | exact 5/5   | exact 5/5  | pass 5/5  |

**Supporting lifecycle and capture checks.** Same-stack cancellation/reclamation checks match all 16 clean rank-geometry cells and reach the expected gate in all 48 preregistered schedule-fault trials (Tables 19–20), extending the registered lifecycle through cancel, scrub, reclaim, lease change, and replacement under the frozen schedule.

**Out-of-process GDN observation.** Across two fresh  $N = 2$  GDN-policy cells and three phases per cell, one spawn-created observer per cell imports live tensor views through PyTorch CUDA IPC. The receiver processes derive 1,080 storage descriptors and all 96,660 pair relations; frozen replay passes 6/6 phase and 2/2 lifecycle verdicts. Both observer processes differ from the producer process, and the worker imports no candidate or model-runtime module and receives no candidate rows, verdicts, or live phase/policy/completion labels. This is process-separated PyTorch/CUDA-IPC GDN observation, not OS/driver ground truth or independent model execution; the producer still enumerates and semantically binds the frozen slots.

**Bounded two-stream lifecycle.** Two host threads on distinct CUDA streams produce positive full-model-call interval overlap both before cancellation (260.802 ms) and after quiescent scrub/reclaim/replacement (95.439 ms). All four tokens and full-logit samples, 20 terminal KV layer pairs, two 60-tensor GDN aggregates, and ownership/lifecycle receipts match the serialized reference. This establishes call-interval overlap and one clean quiescent transition, not simultaneous kernels or arbitrary in-flight cancellation.

### 5.3 SEPARATE DEPLOYMENT STORE-F1 MEASUREMENT

This separate panel reports retained document state and mean F1, not FORKAUDIT validation. In the Transformers 5.14.1 block, full-prefix Q16 retains 140.34 MiB/document; Q8 and per-layer mixed retain 15.89 and 9.74 MiB—88.68%/93.06% less—with mean-F1 deltas of 0.000/−0.022. Full-prefix is faster, so the benefit is Store-F1 rather than latency. The unpooled SGLang/HYPIC block (Liu et al., 2026c) uses the same checkpoint and slice under a different harness; its TP=1 adaptation is not a reproduction of the published TP=2 setting.

Table 3: Same-checkpoint H20 retained Store and mean F1 on an eight-item Qasper/2WikiMQA slice. All rows use Qwen3.5-35B-A3B, one H20-3e per item, a 4,096-token cap, greedy decoding, and at most 32 generated tokens. The Transformers (three repeats/item) and HYPIC (Liu et al., 2026c) blocks are unpooled; HYPIC timing/F1 and Store use separate 24- and 16-cell TP=1 cohorts. Timing is compared only within blocks; no capacity, broad-quality, or ForkAudit-ownership claim is made.

| Method                                                                                 | Persistent state   | Store (MiB)  | TTFT (s) | TPOT (ms) | tok/s | F1    |
|----------------------------------------------------------------------------------------|--------------------|--------------|----------|-----------|-------|-------|
| <i>Transformers 5.14.1 in-process deployment cohort</i>                                |                    |              |          |           |       |       |
| Vanilla dense                                                                          | —                  | —            | 0.649    | 648.75    | 1.36  | 39.42 |
| Full-prefix KV cache                                                                   | Q16                | 140.34       | 0.163    | 54.11     | 13.38 | 39.14 |
| CoMem state                                                                            | Q8                 | <b>15.89</b> | 0.673    | 55.44     | 7.15  | 39.14 |
| CoMem mixed state                                                                      | Q4/Q4/Q8           | 10.01        | 0.674    | 54.97     | 7.19  | 39.01 |
| CoMem per-layer mixed                                                                  | mixed              | <b>9.74</b>  | 0.673    | 55.07     | 7.19  | 39.11 |
| CoMem state                                                                            | Q4                 | 8.41         | 0.674    | 55.02     | 7.19  | 36.09 |
| <i>HYPIC commit 98147c0 timing/F1; receipt-instrumented Store; SGLang 0.5.14, TP=1</i> |                    |              |          |           |       |       |
| Official HYPIC code: full recompute                                                    | —                  | —            | 0.217    | 49.21     | 14.77 | 39.14 |
| Official HYPIC code: prefix cache                                                      | Radix prefix       | 139.53       | 0.072    | 50.15     | 19.07 | 39.14 |
| HYPIC (transition + seam 8)                                                            | segment transition | 324.09       | 0.101    | 52.13     | 17.67 | 39.63 |

Bold marks the headline Store-F1 points. Relative to full-prefix Q16, Q8/per-layer mixed reduce Store 88.68%/93.06%; unrounded mean-F1 deltas are 0.000/−0.022 (39.115 vs. 39.137 for mixed). Per-layer mixed uses Q4 residual, Q16 projections, and layer bits [8, 8, 4, 4, 8, 8, 8]. Store is median retained-document tensor payload and excludes metadata, pools, process deltas, and capacity. Published HYPIC panels use TP=2; this adaptation uses TP=1.

## 6 DISCUSSION AND LIMITATIONS

Within the Section 3 TCB, FORKAUDIT connects canonical equality of registered observables to phase-indexed ownership evidence. The historical case shows why output and terminal-state differentials can miss a persistent-base alias; process-separated reconstruction still trusts producer enumeration.

The 54.5% post-priming allocator result and the separate 88.68%/93.06% retained-Store results use different denominators and cohorts; they are neither additive nor process-memory or capacity results.

Evidence is limited to one Qwen3.5/H20 stack, bounded schedules, captured-boundary checks, and Python-call-scope dispatch. The two-stream cohort shows call-interval rather than kernel overlap and excludes in-flight cancellation. Full capture is costly and targets offline debugging or CI. The constructed controls and one historical mechanism do not estimate population error rates; Appendices H and A state the full boundary.

## 7 CONCLUSION

On one fixed Qwen3.5/H20 stack, FORKAUDIT gives fail-closed, replayable hybrid KV/GDN ownership checks: six targets pass across 96 configurations, while compiled dispatch remains partial. It rejects 44 seeded operator controls, nine all-gates-on mutations, and five separately frozen mutations, and repairs one historical alias in 8/8 cells. Shared-document KV reduces the  $N = 32$  post-priming allocator delta 54.5% with canonical equality of registered observables. Separately, the eight-item Transformers cohort records 88.68%/93.06% Store reductions at 0.000/−0.022 mean-F1 deltas (Table 3). Claims remain conditional on the stated TCB, observation points, and denominators.

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## A REPRODUCIBILITY STATEMENT

Manifest-first CPU replay covers the primary-factorial package and named replayable supporting cohorts, not every contextual table. The primary package rechecks 536 artifacts, 96 cells, 288 cross- $N$  comparisons, eight attention rows, and nine fault pairs. The target-gate-suppression package rechecks 18 cases and 24 FP32 sidecars; its disclosed wrapper changes only an inherited execution-identifier source and preserves every candidate byte. The out-of-process observer package rechecks six IPC captures, 1,080 receiver-derived rows, 96,660 relations, both observer PIDs, and all six phase/two lifecycle verdicts. The bounded two-stream replay recomputes four full-logit comparisons and both interval overlaps.

The designer-executor-separated package binds the PDF-only author freeze, a lifecycle-amended executor, five clean/mutant pairs, detached clean and pair replays, and 208 terminal files. All five clean gates and expected-first classifications pass; the pre-injection amendment was frozen before all five ranks were rerun in a new output root.

The historical-regression package binds the pre-discovery chronology, frozen pre-fix/repair sources, outcome-blind protocol and resource amendment, eight three-lane rank artifacts, 24 FP32-logit sidecars, and 24 fresh nonces. A fresh candidate-import-free replay of every rank and the aggregate is byte-identical to the archived output. The earlier invalid attempt supplies chronology only; none of its outcomes is imported.

The expanded numerical-sweep package binds 272 sidecars totaling 140,199,936 bytes and rechecks 20 attention and 24 GDN clean rows plus 44 controls. HYPIC binds 4,404 timing artifacts and hash-replays all 16 Store cells. No package independently re-executes the model, enumerates OS/driver allocations, or attests compiled dispatch; Appendix I maps exact artifacts and boundaries.

## B ETHICS STATEMENT

This work introduces no human-subject data collection and evaluates only systems behavior on the PG-19 training split. It does not assess model safety, bias, privacy, or downstream deployment quality. Memory efficiency can lower deployment cost but may also increase the scale of model use; those broader effects are outside this controlled systems study.

## C PROTOCOL GEOMETRY

Table 4: Frozen primary-factorial execution geometry. The eight ranks provide distinct books and query banks, not repeated stochastic measurements.

| Component             | Frozen value                                                                                                                                                                                                                                                                     |
|-----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Model                 | Qwen3.5-35B-A3B; 10 attention + 30 GDN layers                                                                                                                                                                                                                                    |
| Hardware              | 8 × H20-3e; one book/rank                                                                                                                                                                                                                                                        |
| Software              | Python 3.11.13; PyTorch 2.11.0+cu129; Transformers 5.14.1; vLLM 0.26.0+cu129; Triton 3.6.0; CUDA 12.9                                                                                                                                                                            |
| KV backend            | vLLM fused attention; BF16/Q16; page 128                                                                                                                                                                                                                                         |
| Document / query      | 4,095 / 32 tokens; PG-19 train only                                                                                                                                                                                                                                              |
| Generation            | 8 output tokens; 39 state-appended input tokens (32 query + 7 generated-token feedback)                                                                                                                                                                                          |
| Resident counts       | 1, 8, 32; four KV-by-GDN ownership cells                                                                                                                                                                                                                                         |
| Primary schedule      | all requests resident; batch-one sequential round-major calls on one CUDA stream/rank                                                                                                                                                                                            |
| Validation            | 96 factor configurations; separate memory/witness cells                                                                                                                                                                                                                          |
| Fault campaigns       | $N = 2$ ; the primary all-gates-on campaign has nine live mutants with rebuilt controls; the target-gate-suppression campaign has nine fresh clean plus nine suppressed-gate cases on 8 H20s; the designer-executor-separated campaign has five new clean/mutant pairs on 5 H20s |
| Historical regression | one previously encountered one-token alias defect; 8 ranks × 3 fresh $N = 2$ lanes; 3 archived-coordinate reproductions + 5 additional frozen inputs; no mutation                                                                                                                |

Table 5: Cohort-to-claim authorization. Cohorts are never pooled; each row supports only the listed inference.

| Cohort                      | Frozen scope                                                                                                                                                                                     | Authorized claim / explicit exclusion                                                                                                                                                                                                                                                                                                                                                  |
|-----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Primary ownership factorial | 8 books; $N = 1, 8, 32$ ; four KV × GDN cells; 8 decode steps                                                                                                                                    | ownership lifecycle, cross-cell equivalence, selected attention oracle, live-mutant sensitivity, and four-cell allocator endpoints; not transfer or scheduler evidence                                                                                                                                                                                                                 |
| Fresh-process control       | one independent $N = 2$ process; one clean path and M1                                                                                                                                           | input reconstruction, cleanup, and one intended-gate reproduction; not pooled with the factorial                                                                                                                                                                                                                                                                                       |
| Lifecycle transfer          | 8 books; aligned 4,096-token prefix; $N = 4$ ; two cancelled slots                                                                                                                               | exact outputs/state, zero-scrub, exact reuse, and stale-lease rejection on the same adapter; not scheduler, concurrency, or performance evidence                                                                                                                                                                                                                                       |
| Scheduler interleave        | 8 books; $N = 4$ ; page/prefix/tail geometries 64/3,985/17 and 128/4,033/65; cancel slot 3 after round 2                                                                                         | deterministic request-step interleaving, zero-scrub, epoch change, exact reuse, uninterrupted-control equality, and three frozen scheduler-fault gates; not concurrent kernels, continuous-batching performance, detection rate, or production end-to-end correctness                                                                                                                  |
| Out-of-process GDN observer | one H20; fixed 4,095-token document and 32-token registered transition; two fresh $N = 2$ shared-KV cells (shared-base/materialized GDN); three phases/cell; one spawn-created observer per cell | 1,080 receiver-derived descriptors and 96,660 pair relations; 6/6 phase and 2/2 lifecycle verdicts pass; no live candidate rows/verdicts or phase/policy/completion labels; producer slot enumeration and PyTorch/CUDA IPC remain trusted; not OS/driver ground truth, independent model execution, KV capture, kernel attestation, transient-write exclusion, or cross-stack evidence |

*Continued on next page*

Table 5 continued.

| Cohort                                     | Frozen scope                                                                                                                                                                                   | Authorized claim / explicit exclusion                                                                                                                                                                                                                                                                                                     |
|--------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Bounded two-stream lifecycle               | one H20; one 4,033-token document; two 16-token query requests at $N = 2$ ; two host threads and distinct CUDA streams; pre-cancel and post-reclaim phases; cancellation after synchronization | positive call-interval overlap plus exact four lifecycle outputs, 20 logical-KV layer pairs, two GDN slot aggregates, ownership, scrub, and reuse; not simultaneous kernels, native continuous batching, in-flight cancellation, production scheduling, or performance                                                                    |
| GDN transition oracle                      | 1 frozen book/window; global model-layer indices 0, 10, 20, 38; 4 query tokens                                                                                                                 | clean recurrence outputs/states and four seeded wrong-transition rejections conditional on post-normalization inputs; not all GDN layers or end-to-end semantics captured-boundary numerical agreement and rejection of all 44 seeded wrong operators; not upstream-activation, independent-capture, full-model, or end-to-end validation |
| Expanded captured-boundary numerical sweep | 2 distinct PG-19 inputs; all 10 attention layers and 12 of 30 GDN layers; 20 attention and 24 GDN clean rows plus one seeded control/row                                                       | per-fault comparison with redundant audit gates, exact allowlisted production assertions, and token/FP32-logit equality; M8's injected sentinel is not a detector; not a blind-fault study or rate                                                                                                                                        |
| Target-gate suppression matrix             | 9 fresh clean + 9 target-suppressed cases; same fixed model/data/ $N = 2$ ; 8 distinct H20s                                                                                                    | all 5 clean gates pass and all 5 mutants fail first at their frozen primary predicate; per-fault sensitivity only, not a detection rate, natural-defect study, or cross-stack evidence                                                                                                                                                    |
| Designer-executor-separated faults         | separate designer reads only the prior PDF; 5 new frozen $N = 2$ clean/mutant pairs; 5 H20s; all gates enabled                                                                                 | pre-fix binding gate and base violation with output/terminal-state equality, plus storage-clean exact repair; one retrospective defect only, not prevalence, recall, statistical replication, or cross-stack evidence                                                                                                                     |
| Historical alias regression                | one pre-discovery integration defect; 3 archived coordinates + 5 additional frozen inputs; fresh pre-fix, repaired, and materialized $N = 2$ one-token lanes; 8 H20s                           | separates split-family agreement from agreement with vanilla dense; motivation only and separate from the H20 case                                                                                                                                                                                                                        |
| Mac common-mode control                    | Apple M4 Pro; six documents/queries; five fresh MLX processes; vanilla dense plus CoMem BF16/Q8/Q4 at depth 7                                                                                  | same-stack retained-Store, TTFT, TPOT, tokens/s, and mean F1 comparison; not pooled with the primary factorial and not scheduler/continuous-batching evidence                                                                                                                                                                             |
| H20 deployment memory-quality              | 8 LongBench validation workloads; vanilla dense, full-prefix KV reuse, and CoMem Q8/Q4/mixed variants (six configurations total) with 3 timing repeats                                         | within-block cache-hit, quality, and client-wall timing context; not pooled with in-process CoMem timings, not a cross-framework leaderboard, and not continuous-batching evidence                                                                                                                                                        |
| Matched serving controls                   | 8 LongBench validation workloads; vLLM 0.26 and SGLang 0.5.17 cache off/on plus 24 fixed-configuration HYPIC timing/quality cells; one H20-3e per workload                                     | median target-entry-owned physical tensor payload only; not timing, NVML/process memory, capacity, continuous batching, or ForkAudit ownership evidence                                                                                                                                                                                   |
| HYPIC retained-state receipts              | 8 Prefix Cache + 8 HYPIC cells on the same fixed TP=1 model/slice; receipt-instrumented SGLang lifecycle                                                                                       | same-checkpoint numerical and logical-state trade-offs only; not an all-layer implementation, LongBench/quality result, speedup claim, or jointly optimized system                                                                                                                                                                        |
| Related-work operator transfer             | Hydragen at $N = 8/32$ on one captured layer-3 attention family; Palu on one layer with 8 calibration and 1 held-out PG19 row                                                                  |                                                                                                                                                                                                                                                                                                                                           |

Table 6: Trace coverage and replay verdict for the seven contract targets in the frozen case. Equality denotes canonical byte-digest identity as recorded. Overall trace coverage remains partial because dispatch lacks a per-call compiled-binary and autotuning-choice binding. A passing primary target verdict is conditional on the primary capture producer and target boundary in Section 3; dispatch has only a Python-call predicate pass and remains partial. The separate out-of-process GDN cohort is supporting evidence and does not independently execute the model or upgrade these seven targets to comprehensive system correctness.

| Target                 | Trace coverage | Replay verdict             | Decisive witness                                                                                                      | Boundary                                                          |
|------------------------|----------------|----------------------------|-----------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------|
| Frozen identity        | complete       | pass                       | identical preflight/terminal source ledgers, terminal model rehash, and stable data/query/protocol/execution bindings | one validated formal run; not independent producer recapture      |
| Prefix immutability    | complete       | pass                       | physical document-KV digests and GDN base guards across all three phases                                              | one frozen configuration/geometry                                 |
| Private ownership      | complete       | pass                       | all-pairs request-base and request-peer range replay at the primary registered 32-token transition                    | pointer-free conservative intervals                               |
| Tail-safe append       | complete       | pass                       | one-time 127-token private-tail copy before append                                                                    | one partial-tail geometry                                         |
| Dispatch provenance    | partial        | Python-call predicate pass | Python callable, query/position/mask/append/scale receipts; four compiled cache artifacts bound                       | no per-call compiled-binary or explicit autotuning-choice binding |
| Cross-arm equivalence  | complete       | pass                       | tokens, shape/dtype-bound CPU-FP32 logit digests, logical KV, and GDN                                                 | one primary sequential schedule                                   |
| Cross- $N$ consistency | complete       | pass                       | 288 all-arm adjacent-fan-out comparisons over 72 rank-query identities                                                | relational, not an independent oracle                             |

Selected attention and GDN numerical oracles, the nine targeted live faults, the lifecycle/scheduler extensions, the out-of-process GDN and bounded two-stream cohorts, the five PDF-only designer-executor-separated faults, and the one historical-regression reproduction are supporting checks rather than additional contract targets. Their captured-input, same-adapter, process-separated-but-shared-IPC, and schedule boundaries are reported separately.

## D POINTER-FREE STORAGE WITNESS SCHEMA

**Target-to-record dependencies.** The following matrix is normative. I, L, E, and F denote the Identity, Lifecycle, Execution, and Target/fault classes in Table 17; parenthesized entries name mandatory fields. Missing any marked field makes that target open. A preregistered geometry may make tail safety not applicable; the absence of a compiled-binary fingerprint leaves dispatch partial rather than silently full. A missing F field invalidates the corresponding fault row, but does not change the separately evaluated seven-target witness.

| Target                         | I                                      | L                                            | E                                                  | F        | Blind predicate                             |
|--------------------------------|----------------------------------------|----------------------------------------------|----------------------------------------------------|----------|---------------------------------------------|
| Frozen identity                | code, model, data, protocol; role map  | —                                            | —                                                  | N/A      | exact bytes and complete roles              |
| Prefix immutability            | document/base start digests            | roles; setup and final guards                | canonical content digests                          | N/A      | start = final                               |
| Private ownership              | request/base/private roles             | setup, registered transition, final bindings | normalized storage ranges                          | N/A      | non-vacuous all-pairs disjointness          |
| Tail-safe append               | page geometry; document/private roles  | first append event                           | ordered append and copied-tail receipt             | N/A      | one private copy before write               |
| Dispatch provenance            | callable and protocol identity         | —                                            | ordered call, query, mask, position, append, scale | N/A      | schedule and callable equality; no fallback |
| Cross-arm equivalence          | factor, query, and schedule identities | final bindings                               | token, logit, logical-KV, GDN records              | N/A      | canonical semantic equality                 |
| Cross- $N$ consistency         | query and fan-out identities           | —                                            | the same canonical semantic records                | N/A      | smallest- $N$ equals each larger $N$        |
| Fault sensitivity (supporting) | frozen target/gate map                 | matched-control rebuild and cleanup          | injected outcome and failed predicates             | required | target, injection, expected gate, outcome   |

Each tensor-view row has the exact fields `storage_id`, `shape`, `stride`, `storage_offset`, `dtype`, `device`, `storage_nbytes`, `tensor_nbytes`, and `content_sha256`; absolute addresses are forbidden. Within each phase, the producer assigns `storage-0000`, `storage-0001`,

and so on in first-observation order to backing storages. Reusing an ID with a different device or storage size is invalid. Phase capture IDs are unique, whereas request and persistent lifecycle-guard IDs must remain stable across setup, the registered transition, and final generation.

Let element size be  $e$ , storage offset  $o$ , shape  $s_i$ , and stride  $t_i$ . The independent replay initializes  $m = M = o$  and, for every dimension, computes

$$d_i = (s_i - 1)t_i, \quad m += \min(d_i, 0), \quad M += \max(d_i, 0). \quad (3)$$

The conservative half-open byte interval is

$$I = [em, e(M + 1)), \quad (4)$$

which must be non-negative and contained in `storage_nbytes`; also `tensor_nbytes` =  $e \prod_i s_i$ . Two rows overlap iff their normalized storage IDs match and  $I_a^{\text{start}} < I_b^{\text{end}}$  and  $I_b^{\text{start}} < I_a^{\text{end}}$ . Exact aliasing additionally requires equality of interval, geometry, dtype/device, storage and tensor sizes, and content digest. This bounding interval can conservatively flag an exotic interleaved view, but it cannot miss a shared byte.

At setup, every recurrent-state coordinate must either exactly alias the persistent state (borrowed-base arm) or be disjoint from it (materialized arm). At the primary registered 32-token transition boundary, a completed request’s 60 mutable rows must be disjoint from the base and every peer; incomplete borrowed requests must remain exact aliases. At final, all request-local GDN states must be disjoint from the persistent GDN base and every peer; document KV remains intentionally shared. Independent binding-token receipts additionally require unchanged tokens for incomplete requests and changed binding/storage tokens for completed requests. The package applies this schema to all 96 timelines and 288 phase artifacts; nine adversarial tests cover exact aliasing, partial overlap, adjacent intervals, negative strides, conflicting ID reuse, pointer injection, capture/guard reuse, and cross-coordinate request–base and request–peer aliasing.

## E EXECUTED FAULT ROWS AND INVARIANT MAP

The row-level campaign is kept out of the main paper because it supports diagnosis and audit coverage rather than a separate headline result. Separately, the CPU governance suite passes 17 tests and rejects 28 explicit serialized-artifact tamper cases plus its registered helper/launcher faults. That result concerns schema and orchestration fail-closure only; it is not GPU semantic or performance evidence.

Table 7: Separate primary all-gates-on reference. Every separately rebuilt matched-clean cell passed; every mutant reached its target gate fixed before execution and was restored before disposal.

| ID | Injected live fault | Expected/observed gate          | Primary outcome     |
|----|---------------------|---------------------------------|---------------------|
| M1 | reservation alias   | KV.RESERVATION.DISJOINT         | target gate reached |
| M2 | sequence swap       | KV.SEQUENCE.ID                  | target gate reached |
| M3 | omit tail COW       | KV.TAIL.COW                     | target gate reached |
| M4 | GDN–base alias      | gdn.completed.vs.base.disjoint  | target gate reached |
| M5 | GDN peer alias      | gdn.completed.vs.peers.disjoint | target gate reached |
| M6 | position off-by-one | POSITION.CANONICAL.VALUES       | target gate reached |
| M7 | materialized mask   | MASK.CONTRACT                   | target gate reached |
| M8 | wrong callable      | KERNEL.CALLABLE.ID              | target gate reached |
| M9 | dense KV view       | KV.PAGED.VIEW                   | target gate reached |

**Designer–executor-separated fault detail.** The five fixed per-fault outcomes summarized in the main text are shown below; they remain constructed controls rather than a defect-population sample.

Table 8: Designer–executor-separated PDF-only fault outcomes. A separate designer saw only the prior 24-page PDF and froze all five definitions, loci, payloads, clean requirements, and expected first predicates before executor preparation or candidate output. The faults are held out from executor preparation and outcome inspection, not from the predicate vocabulary. All matched clean cases pass; every mutant completes its registered horizon and is rejected first by the frozen primary predicate. Semantic comparisons are secondary: “Same” means canonical exact equality with the matched clean case, KV/GDN denote terminal logical state, and “N/C” means full-logit call cardinality is not comparable. These five fixed outcomes are not a population detection rate.

| Frozen fault                 | Isolated mutation                                               | First failed predicate          | Tokens / logits / KV / GDN            |
|------------------------------|-----------------------------------------------------------------|---------------------------------|---------------------------------------|
| HF01 delayed tail detach     | append write precedes the otherwise present tail copy           | tail-copy-before-first-write    | Same / Same / Same / Same             |
| HF02 inactive-lane scribble  | one bit in the unused lane of a shared physical K page          | physical-prefix immutability    | Same / Same / Same / Same             |
| HF03 duplicate commit        | one correct feedback call commits twice; first output discarded | ordered-call cardinality        | Same / N/C / Changed / Changed        |
| HF04 effective-scale drift   | first registered attention scale is exactly doubled             | attention effective scale       | Changed / Changed / Changed / Changed |
| HF05 stale GDN binding token | storage correctly rebinds; lifecycle token remains stale        | completed binding-token advance | Same / Same / Same / Same             |

Table 9: Representative failures each target is designed to expose. These are the nine executed primary-factorial live mutants; the checks are not complete oracles.

| Target                   | Intended sensitivity and observed boundary                                      |
|--------------------------|---------------------------------------------------------------------------------|
| Frozen identity          | wrong model/data/query bank, changed protocol, stale shard (schema adversaries) |
| Prefix immutability      | M3 omits partial-tail COW; physical document digest and ownership gates fail    |
| Private ownership        | M1 reservation alias, M4 request–base GDN alias, M5 request–peer GDN alias      |
| Sequence/view binding    | M2 sequence swap and M9 dense-KV replacement                                    |
| Dispatch provenance      | M6 position offset, M7 materialized mask, M8 wrong callable                     |
| Factorial exactness      | behavior changes when either KV or GDN setup ownership changes                  |
| Bounded numerical oracle | numerical error in a selected fused full-attention result                       |
| Raw replay               | internally inconsistent trajectory, storage, allocator, or aggregate fields     |

The primary all-gates-on execution is a first-gate localization study. Every registered gate fires before the injected path produces a post-injection token, logit, logical-KV digest, or GDN digest; its output cells are therefore N/O rather than misses. Lifecycle/ownership/storage receipts supply the first gate for M1 and M3–M5; request-binding/call receipts supply it for M2 and M6–M9. The full audit rejects all nine at their predeclared gates under passing matched controls. The separate target-gate-suppression matrix then suppresses one named gate per mutant while leaving the remaining observers active. It provides the per-fault comparison in Table 1, but does not estimate a population rate, held-out-fault coverage, or false-positive rate.

## F RELATION TO PRIOR WORK BY AUDIT OBLIGATION

Table 10: Primary-source positioning. “Evidence focus” describes what the cited work evaluates; it is not a claim that the work lacks all other checks.

| Work                  | Primary mechanism                                                                                           | Evidence focus / relation to FORKAUDIT                                                                                                                                                                                                                                                                                                          |
|-----------------------|-------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| PagedAttention        | Paged KV allocation and sharing (Kwon et al., 2023)                                                         | Serving throughput and memory utilization; supplies the block-sharing substrate.                                                                                                                                                                                                                                                                |
| SGLang / Prompt Cache | Prefix reuse in structured or modular prompts (Zheng et al., 2024; Gim et al., 2024)                        | Runtime performance and output behavior; establish reuse and positional handling as prior art.                                                                                                                                                                                                                                                  |
| CoMem                 | One persistent residual per token at split $j$ ; bounded selection; suffix replay (Liu et al., 2026a;b)     | Motivates the reusable document object; the bounded deployment table reports a same-stack retained-memory–quality comparison, while the core contribution here remains the hybrid-state ownership audit.                                                                                                                                        |
| Marconi               | Admission and eviction for hybrid-model prefixes (Pan et al., 2025)                                         | Hybrid cache hit rate and time to first token; closest state-heterogeneous caching system, but a different policy-level contribution.                                                                                                                                                                                                           |
| SCBench               | Lifecycle-oriented KV-cache benchmark (Li et al., 2025)                                                     | Breadth over tasks and cache phases; exposes breadth absent from our single reference case.                                                                                                                                                                                                                                                     |
| MT-DLComp / CRADLE    | Metamorphic compiler testing and cross-backend differential checking (Xiao et al., 2022; Pham et al., 2019) | Direct testing precedents; our audit adds ownership witnesses and a layerwise dense oracle, but not an independent end-to-end backend.                                                                                                                                                                                                          |
| DNNMem                | Component-wise GPU-memory accounting (Gao et al., 2020)                                                     | Supports denominator separation; it is an estimator, not a request-ownership audit.                                                                                                                                                                                                                                                             |
| FORKAUDIT             | Typed, phase-indexed ownership/call/accounting trace with explicit coverage semantics                       | Integrates a mutation-tested KV-by-GDN ownership factorial, captured-boundary attention/GDN numerical sweeps, same-adapter lifecycle/scheduler-contract checks, and bounded out-of-process GDN storage observation; no new cache policy, production scheduler, OS/driver allocation monitor, independent model execution, or end-to-end oracle. |

Table 11: Closest established precedent for each audit obligation. The final column states the narrower integration contributed by FORKAUDIT; it does not claim that any individual mechanism is new.

| Audit obligation                   | Closest established precedent                                                                                         | Integrated obligation in this work                                                                                                                      |
|------------------------------------|-----------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| Frozen identity                    | Controlled variants in metamorphic and differential testing (Chen et al., 2018; Xiao et al., 2022; Pham et al., 2019) | Bind model, data, query bank, callable contract, and protocol bytes to every ownership receipt.                                                         |
| Document immutability and tail COW | PagedAttention block sharing and copy-on-write for shared prefixes (Kwon et al., 2023)                                | Replay physical document pages, padding, active tables, and the partial-tail detach event.                                                              |
| Recurrent-state ownership          | Hybrid recurrent states and their in-place update constraint (Yang et al., 2025; Pan et al., 2025)                    | Witness request–base and request–peer ranges before and after the frozen schedule’s registered transition boundary.                                     |
| Dispatch provenance                | Cross-backend differential checking exposes implementation-dependent behavior (Pham et al., 2019)                     | Hold the backend fixed and record callable, position values, mask, scale, and append history per call.                                                  |
| Cross-arm and cross- $N$ relations | Metamorphic relations over semantics-preserving transformations (Chen et al., 2018; Xiao et al., 2022)                | Treat ownership policy and resident fan-out as transformations that must preserve registered request observables.                                       |
| Bounded numerical oracle           | Independent backend comparison (Pham et al., 2019)                                                                    | Reconstruct selected attention inputs for dense IEEE-FP32 math and selected post-normalization GDN inputs for a candidate-import-free NumPy recurrence. |
| Memory denominators                | Component-wise GPU-memory accounting (Gao et al., 2020)                                                               | Keep page geometry, allocator current/peak, and unmeasured process/service capacity as separate claim domains.                                          |

Table 12: Unpooled Mac motivation context (Apple M4 Pro, five fresh MLX processes; median over 30 query executions at split depth 7). Vanilla dense is the external reference. CoMem BF16/Q8/Q4 variants use the same selected documents. Store size is the six-document residual corpus; online time includes query Write, dequantization, and suffix Read. This small teacher-forced demonstration set is not primary ForkAudit evidence and supports no cross-hardware or production-speed claim.

| Method               | Store (MiB) | Online (ms) | Speedup | Top-1 vs. dense |
|----------------------|-------------|-------------|---------|-----------------|
| Vanilla dense prompt | –           | 194.70      | 1.00×   | 100.00%         |
| CoMem BF16 split     | 1.998       | 168.88      | 1.15×   | 71.89%          |
| CoMem Q8 split       | 1.061       | 169.17      | 1.15×   | 71.89%          |
| CoMem Q4 split       | 0.562       | 169.25      | 1.15×   | 71.89%          |

Q8/Q4 retain 100% top-1 agreement with the BF16 split path, yet all split rows inherit only 71.89% agreement with vanilla dense. The speed column is a small teacher-forced mechanism benchmark, not production RAG throughput.

## G SUPPORTING CONTROL EXPERIMENTS

**Mac common-mode motivation.** The archival Apple-M4-Pro control is kept unpooled from the H20 experiments and serves only to show why split-family agreement does not replace a dense or independently implemented oracle.

**Published systems context.** Table 13 reports each cited system only in its native protocol and keeps all non-commensurate metrics outside our matched panels. This unpooled table is literature context, not ForkAudit evidence.

Table 13: Unpooled published-systems context in each paper’s native protocol. These values show the scale and evaluation breadth of related systems, *not* a cross-paper leaderboard or ForkAudit evidence: models, hardware, batching, baselines, and metrics differ. Consequently, no row is merged with our same-stack H20 Table 3, and the absence of a common score must not be read as a negative result.

| System                                       | Native method and study setting                                                                                                                                               | Reported efficiency in its native protocol                                                       | Quality and comparability boundary                                                                         |
|----------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|
| PagedAttention / vLLM (Kwon et al., 2023)    | Paged KV allocation plus serving engine<br><i>Setting</i> : LLaMA-7B on A10G and LLaMA-13B on A100-40GB; vLLM serving engine                                                  | 2–4× serving throughput at similar latency vs. FasterTransformer/Orca                            | No common LongBench F1; exact cache-management method                                                      |
| Prompt Cache (Gim et al., 2024)              | Modular attention-state reuse<br><i>Setting</i> : Llama2/MPT/Falcon; RTX 4090, A40, and A100; HuggingFace-style prototype                                                     | 1.5–3.1× TTFT (CPU-resident modules); 3.7–11.7× (GPU-resident) on its 12-dataset LongBench suite | LongBench accuracy evaluated, but with different models and hardware                                       |
| SGLang / RadixAttention (Zheng et al., 2024) | Prefix reuse plus structured-program runtime<br><i>Setting</i> : Llama-7B/Mixtral-8x7B and structured programs; A10G/A100; SGLang                                             | Up to 6.4× throughput vs. prior inference systems                                                | Application-suite result; no common LongBench F1                                                           |
| ChunkAttention (Ye et al., 2024)             | Prefix-aware KV chunks and attention kernel<br><i>Setting</i> : attention microbenchmarks; mainly A100-80GB, plus A100-40GB/RTX 4090; CUDA 11.8                               | 3.2–4.8× attention-kernel speedup for 1K–4K shared prompts                                       | Kernel result; no end-to-end common task score                                                             |
| Hydragen (Juravsky et al., 2024)             | Exact batched shared-prefix attention<br><i>Setting</i> : CodeLlama-13B/34B shared-prefix batches; A100 (with H100/L40S microbenchmarks); custom HuggingFace runtime          | Up to 32× CodeLlama-13B end-to-end throughput in large shared-prefix batches                     | Exact attention construction; different model and batching regime                                          |
| Palu (Chang et al., 2025)                    | Low-rank latent KV with reconstruction/fusion<br><i>Setting</i> : Llama/Mistral/LongChat quality studies; Llama-2-7B latency on one RTX 4090; HuggingFace plus custom kernels | 50% KV compression and up to 1.89× RoPE-attention speedup; up to 2.91× with quantization         | Llama/Mistral/LongChat native study; our bounded Qwen3.5 operator transfer is reported separately          |
| Preble (Srivatsa et al., 2025)               | Distributed prompt-sharing scheduling<br><i>Setting</i> : Mistral-7B/Llama-3-70B request traces; two–eight GPUs across A6000/H100 clusters; distributed serving               | 1.5–14.5× average-latency and 2–10× p99-latency improvement                                      | Distributed serving traces; no common task score                                                           |
| Marconi (Pan et al., 2025)                   | Admission/eviction for hybrid-model prefixes<br><i>Setting</i> : Attention–Mamba2 7B geometry; LMSys/ShareGPT/SWE-bench traces; CloudLab policy simulation                    | Up to 34.4× token-hit rate and up to 71.1% or 617 ms lower TTFT                                  | Hybrid models and trace-driven policy study; no common LongBench F1                                        |
| HYPIC (Liu et al., 2026c)                    | Position-independent caching for hybrid full/linear-attention models<br><i>Setting</i> : Qwen3.5-35B-A3B main panels at TP=2 on 8 H20-3e; SGLang 0.5.14                       | 3.25× lower TTFT on average and 1.66× higher QPS over Prefix Cache                               | Four models/five workloads; our same-slice TP=1 timing and retained-state controls are reported separately |

**Same-checkpoint transfer detail.** Table 14 is the matched complement to the published context table: it uses our frozen Qwen3.5 checkpoint and H20-3e, binds the official Hydragen and Palu source revisions, and independently replays all raw operator sidecars. Because it does not execute all layers or LongBench, it is kept separate from Table 3.

Table 14: Bounded same-checkpoint H20 transfers. Hydragen uses frozen Qwen3.5 layer-3 post-RoPE attention; Palu uses eight 256-token PG19 calibration rows and one disjoint held-out row. Errors are relative  $\ell_2$  against the matched uncompressed operator; state ratios are logical selected-operator ratios, not process memory or end-to-end results.

| Port          | Setting       | Matched numerical result                     | Logical state result                    | Timing / boundary                |
|---------------|---------------|----------------------------------------------|-----------------------------------------|----------------------------------|
| Hydragen port | $N = 8$       | Hydragen-dense rel. $\ell_2 = 0.00255$       | 8.50 vs. 64.48 MiB (7.59× smaller)      | 0.287 vs. 0.0965 ms; no speedup  |
| Hydragen port | $N = 32$      | Hydragen-dense rel. $\ell_2 = 0.00262$       | 10.00 vs. 257.94 MiB (25.80× smaller)   | 0.290 vs. 0.259 ms; no speedup   |
| Palu port     | rank 64/head  | held-out K/V 0.487/0.549 (plain 0.711/0.839) | 512 vs. 2,048 B/token (4.00× smaller)   | projection only; no fused kernel |
| Palu port     | rank 128/head | held-out K/V 0.317/0.364 (plain 0.543/0.668) | 1,024 vs. 2,048 B/token (2.00× smaller) | projection only; no fused kernel |
| Palu port     | rank 192/head | held-out K/V 0.149/0.194 (plain 0.341/0.459) | 1,536 vs. 2,048 B/token (1.33× smaller) | projection only; no fused kernel |

Hydragen avoids prefix replication but has no speedup here. Palu improves over plain SVD, but residual error forbids an end-to-end quality claim. Neither is a jointly optimized combined system.

**Matched serving-framework and HYPIC context.** Table 15 gives separately timed vLLM and SGLang cache-off/on controls plus the official-code HYPIC Full Recompute, Prefix Cache, and full HYPIC arms under the shared Qwen3.5 streaming boundary. It remains a timing/quality-only panel and authorizes only within-framework interpretation; the separate 16-cell retained-state receipt cohort appears in Table 3. vLLM and SGLang each record 8/8 prefix hits and preserve 8/8 cache-off predictions. Within its TP=1 block, HYPIC gives 39.63, 0.101 s, and 17.67 tokens/s; published Qwen3.5 panels use TP=2, so this is an adaptation. Its 24 timing cells bind a 4,404-entry terminal artifact ledger. Across the 16 retained-state cells, Prefix Cache retains a median 139.53 MiB and HYPIC retains 324.09 MiB; Prefix Cache and HYPIC retain 8.78× and 20.39× as much Store as CoMem Q8, respectively, under the separate Store estimand. This remains single-stream; it does not measure continuous batching.

Table 15: Unpooled timing/quality-only serving context on Qwen3.5 and the same eight-item Qasper/2WikiMQA slice. Each row uses one H20-3e per item, a 4,096-token cap, greedy decoding, and at most 32 generated tokens. Times are OpenAI-compatible streaming client wall-clock; F1 is the eight-item mean. Hit is interpreted only against the disabled/full-recompute reference inside each framework block. No cross-framework speedup, retained-state, scheduler-throughput, capacity, or broad-quality claim is made.

| Framework     | Prefix reuse      | F1    | TTFT (s) | TPOT (ms) | tok/s | Hit |
|---------------|-------------------|-------|----------|-----------|-------|-----|
| vLLM 0.26     | off               | 39.63 | 1.445    | 56.31     | 4.70  | –   |
| vLLM 0.26     | align on          | 39.63 | 0.179    | 53.99     | 14.00 | 8/8 |
| SGLang 0.5.17 | off               | 39.14 | 0.283    | 53.38     | 12.73 | –   |
| SGLang 0.5.17 | Radix on          | 39.14 | 0.146    | 53.15     | 15.83 | 8/8 |
| HYPIC code    | full recompute    | 39.14 | 0.217    | 49.21     | 14.77 | –   |
| HYPIC code    | prefix cache      | 39.14 | 0.072    | 50.15     | 19.07 | 8/8 |
| HYPIC         | transition+seam 8 | 39.63 | 0.101    | 52.13     | 17.67 | 8/8 |

vLLM uses experimental Mamba align; SGLang 0.5.17 uses `extra_buffer` Radix caching. The HYPIC rows use the authors’ commit 98147c0 on SGLang 0.5.14, TP=1, full `transition.rope.recompute`, and an eight-token seam. Its full and prefix rows are controls from that same codebase; published TP=2 results and the separate retained-state receipt cohort are not inserted here.

**Hybrid-prefix policy context.** The separate preregistered Marconi trace extends the eight frozen workloads with a fixed synthetic follow-up turn and evaluates only prefix-policy token reuse under the artifact’s native Attention-Mamba2 geometry. It neither reruns LongBench quality nor supplies Qwen3.5 serving speed, so it is not pooled with the preceding panel or the primary audit.

Table 16: Unpooled preregistered policy-simulator context on a 128-event synthetic multi-turn extension of the eight frozen workloads. Values are token-hit percentages under the official Marconi artifact’s native Attention–Mamba2 geometry. The wrapper calls each policy’s own eviction routine after insertion to close actual tree size to the declared budget. These are not Qwen3.5 quality, latency, or throughput measurements and do not validate the primary ownership audit.

| Policy  | 5 GB         | 10 GB | 20 GB |
|---------|--------------|-------|-------|
| vLLM+   | 2.07         | 6.18  | 46.83 |
| SGLang+ | 90.78        | 93.86 | 93.86 |
| Marconi | <b>93.86</b> | 93.86 | 93.86 |

Table 17: Normative record classes. “Mandatory” means that absence makes the corresponding target open rather than silently not applicable.

| Record class | Mandatory payload                                                         | Independent replay                            |
|--------------|---------------------------------------------------------------------------|-----------------------------------------------|
| Identity     | frozen code, model, data, protocol, and object-to-role map                | exact bytes and role coverage                 |
| Lifecycle    | setup, registered-transition, and final inventories and bindings          | normalized ranges, aliases, and disjointness  |
| Execution    | ordered call/append receipts, semantic records, named allocator snapshots | frozen schedule, digests, and phase equations |
| Target/fault | predeclared target, gate, matched control, injection, and outcome         | failed-predicate set and classified outcome   |

**Primary allocator endpoints.** Table 18 keeps the four ownership cells and their allocator denominators together under the primary 8-rank cohort.

Table 18: Primary  $8 \times H20-3e$  allocator results at  $N = 32$  (median across ranks; rank values coincide). Vanilla full-copy is the external control; the paged-prefix row is the closest same-stack prefix-sharing baseline. The remaining rows isolate recurrent-base ownership. All cells retain the persistent document source. Values are GiB above the frozen post-priming baseline; generation increment is the additional peak above allocation present at generation start.

| Empirical arm          | KV / GDN setup                        | Final | Peak  | Gen. incr. |
|------------------------|---------------------------------------|-------|-------|------------|
| Vanilla full-copy      | Full-copy KV / Materialized GDN base  | 4.901 | 4.920 | 0.019      |
| GDN ownership ablation | Full-copy KV / Borrowed GDN base      | 4.890 | 4.907 | 1.951      |
| Paged-prefix baseline  | Shared-doc KV / Materialized GDN base | 2.229 | 2.843 | 0.019      |
| Audited hybrid fork    | Shared-doc KV / Borrowed GDN base     | 2.229 | 2.843 | 1.950      |

**Aligned cancellation and reclamation lifecycle.** The separate lifecycle-transfer cohort uses eight ranks, eight distinct PG-19 training books, an aligned 4,096-token prefix, and  $N = 4$ . It is not pooled with the primary factorial.

Table 19: Lifecycle-transfer obligations on the existing Qwen3.5/vLLM Q16 adapter. Counts are deterministic rank outcomes, not stochastic replicates.

| Obligation                                                            | Result                          |
|-----------------------------------------------------------------------|---------------------------------|
| Frozen static/code/model/weight/raw bindings                          | pass; 14/14 weights, 8/8 shards |
| Full-vocabulary logits and generated tokens vs. uninterrupted control | exact, 8/8                      |
| Final logical KV and final GDN state                                  | exact, 8/8                      |
| Aligned prefix, immutable document, and zero partial-tail copy        | pass, 8/8                       |
| Cancelled-slot zero-scrub and exact reservation reassignment          | pass, 8/8                       |
| Stale epoch-0 rejection and matched epoch-1 acceptance                | 8/8; 0 escapes, 0 wrong gates   |

**Scheduler-managed request-step interleaving.** The separate formal extension keeps four requests resident, cancels slot 3 after its second round, zero-scrubs and increments its lease epoch, reassigns the exact reservation, and completes the replacement interleaved with the three survivors. It is not pooled with the primary factorial or the aligned lifecycle-transfer cohort.

Table 20: Formal scheduler-managed request-step interleaving on the frozen Qwen3.5/ForkAudit vLLM-Q16 stack. Clean counts are deterministic rank outcomes; fault counts are preregistered expected-gate outcomes, not a detection rate.

| Case             | Frozen geometry or intervention                                                            | Outcome                                                                    |
|------------------|--------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|
| Clean geometry 1 | page 64; 3,985-token prefix; 17-token document tail; 18-event cancel/replacement lifecycle | all semantic/storage gates and uninterrupted-control comparisons pass, 8/8 |
| Clean geometry 2 | page 128; 4,033-token prefix; 65-token document tail; same lifecycle                       | all semantic/storage gates and uninterrupted-control comparisons pass, 8/8 |
| FH1              | dispatch cancelled epoch-0 lease                                                           | STALE_SLOT_LEASE, 16/16                                                    |
| FH2              | dispatch lease under another request                                                       | LEASE_REQUEST_MISMATCH, 16/16                                              |
| FH3              | reclaim before zero scrub                                                                  | RECLAIM_NOT_ZERO, 16/16                                                    |

**Candidate-import-free GDN transition oracle.** The oracle uses one frozen PG-19 window and four query-time GDN transitions at global model-layer indices 0, 10, 20, and 38. A single preregistration frozen before the successful run binds the post-native-q/k-normalization boundary, scale  $1/\sqrt{128}$ , tolerances, and seeded faults. Its NumPy reference starts at that captured boundary; no candidate module is imported by the reference.

Table 21: PDF-visible oracle selection and replay boundary. Every listed row was locked before its successful candidate execution; neither oracle validates upstream activation construction or end-to-end model behavior.

| Oracle    | Frozen rows / selection rule                                                                                                                                   | Captured boundary                                                   | Decision rule                                                                           | Replay availability                                                       |
|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------|-----------------------------------------------------------------------------------------|---------------------------------------------------------------------------|
| Attention | rank $r$ uses layer $4r + 3$ and round $r$ , for $r \in \{0, \dots, 7\}$ ; request 0, $N = 1$ , shared-document KV + borrowed GDN; frozen rank/layer/round map | post-RoPE Q/K/V, positions, mask, scale, and ordered append shadows | dense CPU-FP32 rel. $L_2 \leq 0.005$                                                    | 8 raw rows and numerical sidecars; primary-package one-command replay     |
| GDN       | global model layers 0, 10, 20, 38; four query tokens from one frozen PG-19 window                                                                              | post-native-q/k-normalization recurrence; scale $1/\sqrt{128}$      | output rel. $L_2 \leq 0.005$ ; state rel. $L_2 \leq 10^{-4}$ ; seeded fault must reject | 4 clean + 4 fault decisions from 40 sidecars; separate one-command replay |

Table 22: Bounded GDN recurrence oracle. Relative errors compare native outputs and final states with the candidate-import-free NumPy FP32 recurrence.

| Global layer | Output rel. $L_2$ | State rel. $L_2$      | Clean | Seeded fault       | Rejected |
|--------------|-------------------|-----------------------|-------|--------------------|----------|
| 0            | 0.001584          | $1.23 \times 10^{-7}$ | yes   | omit decay         | yes      |
| 10           | 0.001652          | $1.36 \times 10^{-7}$ | yes   | complement $\beta$ | yes      |
| 20           | 0.001572          | $1.39 \times 10^{-7}$ | yes   | pre-decay memory   | yes      |
| 38           | 0.001638          | $1.30 \times 10^{-7}$ | yes   | roll value heads   | yes      |

Table 23: Preregistered expanded captured-boundary numerical sweep. Clean rows use candidate-import-free CPU/NumPy FP32 replay; every clean row has one seeded wrong-operator control. Attention begins after RoPE and GDN begins after native q/k normalization, so the table does not validate upstream activation construction, independent capture, or end-to-end model behavior.

| Operator  | Inputs | Layers | Clean rows | Positions / transitions | Max output rel. $L_2$                   | Controls rejected |
|-----------|--------|--------|------------|-------------------------|-----------------------------------------|-------------------|
| Attention | 2      | 10/10  | 20/20      | 160 positions           | 0.001897                                | 20/20             |
| GDN       | 2      | 12/30  | 24/24      | 192 transitions         | 0.002073 ( $2.08 \times 10^{-7}$ state) | 24/24             |

**Full target-gate-suppression matrix.** The preregistered target-gate-suppression execution contains all M1–M9 faults and a fresh matched clean path for each. Across eight distinct H20s, all 18 cases complete their declared horizon or an allowed classified stop, all 24 FP32 sidecars verify, and no case is operationally invalid. Five mutant paths complete semantics: token equality catches 0/5, exact logits catch M5 only ( $L_\infty = 0.75$ , relative  $L_2 = 0.0314$ ), and M1/M2/M6/M7 leave both exact. M3/M4 reach other audit gates; M9 reaches the paired-view production assertion; M8 reaches the exact injected payload sentinel, which is not a production detector. The frozen aggregator initially read the inherited execution identifier from a detached-manifest field that the manifest schema omits. A disclosed postexecution wrapper derives that identifier from the canonical receipt embedded in all eight manifest-bound primary-package shards, changes only the generated comparison-row `run_id` from null to the preregistered value, and reproduces the summary byte-for-byte without modifying candidate outputs or frozen sources.

## H EXPANDED LIMITATIONS AND CLAIM BOUNDARY

The primary factorial covers one model, one H20 hardware family, one page size, 16-bit KV, and  $N \leq 32$ . Its ownership conclusion is tied to one 4,095-token partial-tail geometry, a registered 32-token continuation block, and the subsequent frozen sequential schedule; it is not a claim about an arbitrary first token or transition length. The lifecycle cohort adds one aligned 4,096-token geometry and a fixed cancel/scrub/reclaim sequence. The scheduler-interleave extension adds two geometries and deterministic request-step interleaving, but those requests execute sequentially on one CUDA stream.

The separate two-stream cohort has two host workers and distinct CUDA streams with positive full-model-call interval overlap. Its intervals include queuing, resource waits, and host-enqueue gaps; there is no per-kernel overlap trace. Cancellation occurs only after synchronization, so the cohort does not test arbitrary or in-flight reclamation. The out-of-process GDN observer removes same-process reconstruction and live judgment labels, but the producer still enumerates and semantically binds the frozen slots; both sides trust PyTorch tensor/storage and CUDA-IPC semantics, and mutation is paused for each snapshot. It does not supply OS/driver allocation ground truth, independent model execution, KV/kernel attestation, or protection against a transient write followed by restoration.

No cohort evaluates native continuous/ragged batching, eviction, multi-document serving, general scheduler cancellation, process-level NVML memory, OOM capacity, or a broad downstream benchmark. Full-copy KV and materialized GDN bases are controlled ownership factors rather than optimized production baselines. The GDN path is the Transformers Torch implementation, not an optimized recurrent kernel. The compiled-cache ledger does not bind each call to a selected binary or expose an explicit autotuning artifact.

The auxiliary serving panel uses one request stream per engine and compares each cache-enabled framework only with its own disabled reference. It does not measure continuous batching, eviction, admission control, maximum concurrency, or best-tuned performance. CUDA graphs were disabled in both pinned runs; the reported values describe that fixed configuration only.

The HYPIC timing/quality comparison uses the authors' exact tracked commit on SGLang 0.5.14 in one fixed CUDA 12.9 / PyTorch 2.11 configuration, adapting Qwen3.5 execution from the paper's TP=2 to TP=1 so every arm uses one H20-3e per workload. Cache-hit authority comes from OpenAI response usage. Each of its 24 timing/quality cells has one formal measurement after a discarded, prefix-disjoint warmup. A separate receipt-instrumented run uses the same model/slice and fixed TP=1 serving configuration to produce 8 Prefix Cache and 8 HYPIC retained-state cells; each completed cell binds raw output plus pre-measurement and terminal ownership receipts, and all 16 pass an external hash-bound replay. The instrumentation adds target-entry byte-range receipts and a lifecycle-idempotency overlay; Store therefore does not come from the unmodified timing run. Neither cohort establishes scheduler throughput, QPS, capacity, the paper's TP=2 latency, or accuracy robustness beyond this eight-item slice.

The same-checkpoint related-work transfer is likewise operator-bounded: Hydragen uses one captured layer-3 shared-prefix attention family, and Palu uses one layer with eight calibration rows and one held-out row. Neither executes all model layers, a fused Palu kernel, LongBench, or a jointly optimized Hydragen/Palu-plus-CoMem serving path.

The page equations are deterministic identities, not learned laws; there is one observation per rank-configuration cell and no stochastic replication. Most runtime tensors are represented by hashes and equality receipts rather than archived in full. The attention and GDN oracle rows are the exception and ship numerical sidecars, but cover only selected captured-input operator boundaries rather than an end-to-end model. The expanded numerical sweep covers two short inputs, all ten attention layers, and 12 of 30 GDN layers; it does not establish length/input robustness or full recurrent-layer coverage. Both the original and expanded GDN runs start after native q/k normalization and therefore do not validate that upstream boundary. Pointer-free normalized intervals plus raw-byte hashes make the primary captured storage evidence replayable. The out-of-process observer reconstructs receiver-visible GDN descriptors and relations in another process, but still trusts producer-side slot selection and the IPC substrate; neither constitutes protection against a malicious producer before capture. The nine mutants are targeted positive controls; the target-suppression matrix adds a same-system comparison for those same designed faults, not blind or naturally occurring defects. Four complete with unchanged measured tokens and exact logits, one changes only full logits, and four stop earlier; N/O cells are not misses, while the injected M8 sentinel is not a production detector. This is not a general detection rate, a blind-fault result, or superiority over every conventional test suite. The scheduler extension's three frozen faults are expected-gate outcomes, not a detection rate or evidence of fault-set completeness. Broader systems claims would additionally require a native production serving scheduler, per-kernel overlap evidence, in-flight cancellation tests, optimized controls, alignment/length/input sweeps, and process-level capacity accounting.

The five additional designer-executor-separated faults are genuinely fresh relative to the disclosed PDF fault list and were fixed before executor preparation, but they remain deliberately constructed, fixed-stack mutations. Passing all five does not estimate unseen-fault recall, false-positive rate, or robustness to arbitrary runtime, model, schedule, or adversarial capture defects.

The retrospective case study concerns one organically encountered integration defect in our borrowed-state path. Only ranks 0-2 reproduce its archived input coordinates; ranks 3-7 are additional frozen inputs, and none is a statistically independent defect. The study does not estimate natural-defect prevalence or recall. Output and terminal-state differentials miss the alias, while a persistent-base content invariant catches it; FORKAUDIT contributes authenticated transition-time owner/layer/family localization rather than exclusive detection over all conventional testing.

## I ARTIFACT AND CLAIM MAP

**Artifact footprint.** The source-complete primary package, derived summaries, raw ledgers, and expanded numerical sidecars are content-bound and locally replayable. The table below maps each

manuscript claim to its machine-readable source and validation boundary; supporting measurements not used by a manuscript claim remain in the internal experiment registry.

Table 24: Artifact and claim map. Relative paths are rooted at the accompanying research package. Completed rows authorize only their stated bounded claims. Metadata-redacted derivatives preserve numerical fields and state digests, while most full tensors still require the pinned environment for independent numerical recomputation.

| Claim/evidence                                           | Machine-readable source                                                                                                                                                                                                                                                                                                                                   |
|----------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Primary factorial replay package                         | evidence/round_04_rr2_package/                                                                                                                                                                                                                                                                                                                            |
| Replay and package manifest                              | evidence/round_04_rr2_package/replay/run_replay.sh and evidence/round_04_rr2_package/MANIFEST.json                                                                                                                                                                                                                                                        |
| Registered, derived, and raw evidence                    | upstream/forkaudit-summary.json, derived/derived_summary_v2.json, and upstream/raw/ under the primary replay package                                                                                                                                                                                                                                      |
| Executed source and storage tests                        | executed_source/gpu/ and replay/test_storage_witness.py under the primary replay package                                                                                                                                                                                                                                                                  |
| Identity, lifecycle, scheduler, and GDN extensions       | evidence/rr2_terminal_closure_a1_20260821/, evidence/lifecycle_transfer_reviewer_package/, evidence/round23_a2_scheduler_interleave/, and evidence/gdn_transition_oracle_preregistered_20260819d/                                                                                                                                                         |
| Mac and H20 deployment controls                          | evidence/mac_m4_motivation/, evidence/h20_deployment_benchmark/, and their generators under scripts/                                                                                                                                                                                                                                                      |
| HYPIC retained-state receipts                            | evidence/related_work_same_protocol/hypic-retained-state-r34-trial1892234/                                                                                                                                                                                                                                                                                |
| Target-gate-suppression preregistration and raw ledger   | evidence/r28_full_detector_matrix/formal_run_20260824a/preregistration/detector-matrix-v2-plan.json evidence/r28_full_detector_matrix/formal_run_20260824a/receipts/raw-artifacts.sha256                                                                                                                                                                  |
| Target-gate-suppression summary and replay receipt       | evidence/r28_full_detector_matrix/formal_run_20260824a/detector-matrix-v2-summary.postexec-corrected.json evidence/r28_full_detector_matrix/formal_run_20260824a/postexecution-correction-replay-receipt.json; disclosed field-source amendment: evidence/r28_full_detector_matrix/postexecution-rr2-run-binding-correction.json                          |
| vLLM, SGLang, HYPIC timing/quality, and Marconi controls | evidence/related_work_same_protocol/; exact package members and one-command replays reside under this directory                                                                                                                                                                                                                                           |
| Hydragen/Palu same-checkpoint transfers                  | evidence/round24_related_work_transfer/ and its executed_source/ directory                                                                                                                                                                                                                                                                                |
| Published context and table generator                    | literature/reported_system_context.json and scripts/generate_related_work_context_table.py                                                                                                                                                                                                                                                                |
| Out-of-process GDN observer                              | evidence/r33_independent_capture/formal_h20/result/preregistration/preregistration.json evidence/r33_independent_capture/formal_h20/result/raw/out-of-process-gdn-capture.json evidence/r33_independent_capture/formal_h20/result/replay/out-of-process-gdn-replay.json evidence/r33_independent_capture/formal_h20/result/receipts/terminal-files.sha256 |

*Continued on next page*

Table 24 continued.

| Claim/evidence                                   | Machine-readable source                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|--------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Designer-executor-separated faults               | evidence/r33_fresh_faults/author_freeze/<br>designer_input/round32_input.pdf<br>evidence/r33_fresh_faults/author_freeze/<br>PROTOCOL.md<br>evidence/r33_fresh_faults/executor_attempt_b/<br>AMENDMENT.md<br>evidence/r33_fresh_faults/formal_h20/RESULT_<br>VERIFICATION.json<br>evidence/r33_fresh_faults/formal_h20/<br>r33-fresh-faults-20260825b/rank-run-*/rank-*/<br>mutant-case.json<br>evidence/r33_fresh_faults/formal_h20/<br>r33-fresh-faults-20260825b/summary.json<br>evidence/r33_fresh_faults/formal_h20/<br>r33-fresh-faults-20260825b/terminal-files.<br>sha256                                  |
| Historical alias regression<br>and repair        | evidence/r35_historical_alias_regression/<br>design_decision.md<br>evidence/r35_historical_alias_regression/<br>preregistration.json<br>evidence/r35_historical_alias_regression/<br>preexecution-freeze-receipt.json<br>evidence/r35_historical_alias_regression/<br>preexecution-resource-amendment.json<br>evidence/r35_historical_alias_regression/<br>formal_h20/RESULT_VERIFICATION.json<br>evidence/r35_historical_alias_regression/<br>formal_h20/r35-historical-alias-20260826a/<br>aggregate.json<br>evidence/r35_historical_alias_regression/<br>formal_h20/r35-historical-alias-20260826a/<br>replay/ |
| Bounded two-stream<br>lifecycle                  | evidence/r29_true_concurrency/formal_run_<br>20260825b/raw/formal-result.json<br>evidence/r29_true_concurrency/formal_run_<br>20260825b/replay/independent-replay.json<br>evidence/r29_true_concurrency/formal_run_<br>20260825b/receipts/raw-and-replay.sha256                                                                                                                                                                                                                                                                                                                                                   |
| Expanded<br>captured-boundary<br>numerical sweep | evidence/r30_expanded_oracle_sweep/; 272 local arrays<br>under raw/sidecars/; canonical result oracle-result.json;<br>local replay source r30_expanded_oracle_reference.py;<br>validation validation_report.json; terminal binding<br>terminal-products.sha256                                                                                                                                                                                                                                                                                                                                                    |
| Manuscript claim index                           | evidence/claim_evidence_map.tsv                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |